Linear algebra preliminaries for Quantum Physics

September 8, 2026

1 Complex vector

1. Complex vector in $\mathbb{C}^n$: ψ.
   Its components:

$$\psi = \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix} \tag{1}$$

2. From now on, by *vector* we mean complex vector.

3. An operation with vectors: *Addition* of two vectors.

$$\begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix} + \begin{bmatrix} z'_1 \\ \vdots \\ z'_n \end{bmatrix} \equiv \begin{bmatrix} z_1 + z'_1 \\ \vdots \\ z_n + z'_n \end{bmatrix} \tag{2}$$

4. Another operation with vectors: *Multiplication* of a vector with a scalar.

$$z\psi = z \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix} = \begin{bmatrix} zz_1 \\ \vdots \\ zz_n \end{bmatrix} \tag{3}$$

5. In quantum physics, we use the *braket notation*, also known as the *Dirac notation*:

$$\psi \equiv |\psi\rangle\,, \tag{4}$$
$$\psi + \psi' \equiv |\psi\rangle + |\psi'\rangle\,, \tag{5}$$
$$z\psi \equiv z\,|\psi\rangle\,. \tag{6}$$

2 Scalar product (or inner product) of two complex vectors

1. *Scalar product* of two vectors:

$$|\psi\rangle = \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix}, \tag{7}$$

$$|\varphi\rangle = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \tag{8}$$

$$(|\psi\rangle, |\varphi\rangle) = \sum_{i=1}^{n} z_i^* x_i. \tag{9}$$

   where z^* stands for complex conjugate of z.

2. Exercise (Python-1/1): Compute the scalar product of two complex vectors.

3. Scalar product in the Dirac notation: $(|\psi\rangle, |\varphi\rangle) \equiv \langle\psi|\varphi\rangle$.

4. The *norm* of a vector $|\psi\rangle$ is $|| \, |\psi\rangle \, || = \sqrt{\langle\psi|\psi\rangle}$.

5. Vector $|\psi\rangle$ is a *unit vector*, if its norm is 1.

   Unit vectors are also called *normalized*.

6. Two vectors ψ and φ are *orthogonal*, if $\langle\psi|\varphi\rangle = 0$.

7. An *orthonormal basis* of $\mathbb{C}^n$ is a list $(\varphi_1, \ldots, \varphi_n)$ of n unit vectors that are pairwise orthogonal to each other.

   They form a linearly independent set, and span $\mathbb{C}^n$.

   An example is the *canonical basis*, where the n-th basis vector has 1 as its n-th vector component and zero as all further components, e.g., in $\mathbb{C}^4$ these are

$$|\varphi_1\rangle = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \; |\varphi_2\rangle = \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \; |\varphi_3\rangle = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \; |\varphi_4\rangle = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}. \tag{10}$$

8. The *dual vector* of a vector $|\psi\rangle$ is a $\mathbb{C}^n \to \mathbb{C}$ linear map which maps a vector $|\chi\rangle$ to the scalar $(|\psi\rangle, |\chi\rangle) \equiv \langle\psi|\chi\rangle$.

   The dual vector of $|\psi\rangle$ is denoted as $\langle\psi|$.

   In quantum terminology, a vector is sometimes called a *ket*, whereas a dual vector is sometimes called a *bra*.

3 Complex matrix

1. Here, *complex matrix* will refer to an $n \times n$ complex matrix, $A \in \mathbb{C}^{n \times n}$.

   An example for $n = 2$:

   $$A = \begin{bmatrix} 1+i & 2 \\ -2i & -4 \end{bmatrix}. \tag{11}$$

   Matrix elements are denoted as A_{ij} with $i, j \in \{1, \ldots, n\}$. In this example: $A_{11} = 1 + i$, $A_{12} = 2$, $A_{21} = -2i$, $A_{22} = -4$.

2. Examples, with $n = 2$.

   Null matrix:

   $$N = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}. \tag{12}$$

   Unit matrix:

   $$\sigma_0 \equiv I \equiv \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}. \tag{13}$$

   Pauli matrices:

   $$\sigma_1 \equiv \sigma_x \equiv X \equiv \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \tag{14}$$

   $$\sigma_2 \equiv \sigma_y \equiv Y \equiv \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}, \tag{15}$$

   $$\sigma_3 \equiv \sigma_z \equiv Z \equiv \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}. \tag{16}$$

   Remark: *Pauli matrices* sometimes mean the set $\{I, X, Z, Y\}$, sometimes only the set $\{X, Y, Z\}$.

3. The *outer product* or *dyadic product* of two vectors

   $$|\psi\rangle = \begin{bmatrix} z_1 \\ \vdots \\ z_n \end{bmatrix}, \text{ and } |\varphi\rangle = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} \tag{17}$$

   is the matrix A with elements

   $$A_{ij} = z_i x_j^*. \tag{18}$$

   This outer product A is often denoted in braket notation as $|\psi\rangle \langle\varphi|$.

4. Exercise (Python-1/2): Compute the dyadic product of two vectors.

4 Eigenvector, eigenvalue

1. The vector $|\psi\rangle$ is an *eigenvector* of matrix A, if $A|\psi\rangle = \lambda|\psi\rangle$ with some scalar $\lambda \in \mathbb{C}$. In this case, λ is called the *eigenvalue* of A corresponding to eigenvector $|\psi\rangle$.

   The collection of all eigenvalue-eigenvector pairs of a matrix is called an *eigensystem*.

2. Eigenvalues of a matrix A can be found by finding the roots of the characteristic polynomial $f(\lambda) = \det(A - \lambda I)$.

3. Exercise (Python-1/3): Compute the eigenvalues and eigenvectors of a matrix.

4. The *diagonal representation* of a matrix A is a representation $A = \sum_{i=1}^{n} \lambda_i |\phi_i\rangle\langle\phi_i|$, where the vectors $|\phi_i\rangle$ form an orthonormal set of n eigenvectors for A, with corresponding eigenvalues λ_i.

5. A matrix is *diagonalizable* if it has a diagonal representation.

6. Example for a diagonalizable matrix: Z. Example for a non-diagonalizable matrix:

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}. \tag{19}$$

7. There can be multiple eigenvectors belonging to a given eigenvalue. In that case, these eigenvectors belonging to the same eigenvalue form a subspace. The dimension of this subspace is the degree of *degeneracy* of that eigenvalue.

   For example, 2 is an eigenvalue of this A with twofold degeneracy:

$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 0 \end{bmatrix}. \tag{20}$$

5 Adjoint of a matrix. Hermitian matrix. Normal matrix. Unitary matrix.

1. The *adjoint* or *Hermitian conjugate* of a matrix $A \in \mathbb{C}^{n\times n}$ is the matrix

$$\left(A^\dagger\right)_{ij} = A^*_{ji}. \tag{21}$$

   That is, the adjoint of A is the transpose of the complex conjugate of A, or the complex conjugate of the transpose of A.

2. The adjoint of the adjoint of a matrix is the matrix itself.

3. A matrix is *self-adjoint* or *Hermitian*, if it is its own adjoint: $A^\dagger = A$.

4. The outer product of a vector with itself, $|\psi\rangle\langle\psi|$, is Hermitian.

5. Given a k-dimensional subspace of $\mathbb{C}^n$, with $1 \leq k \leq n$, and the orthonormal basis $\{|\phi_1\rangle, \ldots |\phi_k\rangle\}$ of this subspace, the *projector* onto that subspace is

$$P = \sum_{i=1}^{k} |\phi_i\rangle \langle\phi_i| . \tag{22}$$

6. All projectors are Hermitian.

7. Any projector P satisfies $P^2 = P$.

8. Exercise (Python-1/4). Formulate the projector of a one-dimensional subspace, given a vector from that subspace.

9. A matrix A is *normal*, if $AA^\dagger = A^\dagger A$.

10. *Spectral decomposition theorem.* A matrix is normal if and only if it is diagonalizable.

11. A normal matrix is Hermitian if and only if it has real eigenvalues.

12. Exercise (easy): Compute the eigenvalues and eigenvectors of all four Pauli matrices.

13. A matrix U is *unitary* if $U^\dagger U = 1$.

14. All eigenvalues of a unitary matrix have unit absolute value.

15. For a unitary matrix, $UU^\dagger = 1$.

16. A unitary matrix preserves the scalar product: for any two vectors $|\psi\rangle$ and $|\varphi\rangle$, and any unitary matrix U, it holds that

$$\langle U\psi|U\varphi\rangle = \langle\psi|\varphi\rangle . \tag{23}$$

17. Exercise (Python-1/5): Compute eigenvalues and eigenvectors of a unitary matrix, and show that the spectral decomposition formula yields the original matrix.

18. Exercise (easy): Show that all four Pauli matrices are Hermitian and unitary.

6 Matrix functions

1. Exercise (Python-1/6): Compute the exponential of a matrix.

2. Take a complex function $f : \mathbb{C} \to \mathbb{C}$.

   This can be generalized as a function that maps $n \times n$ diagonalizable matrices to $n \times n$ diagonalizable matrices, in the following way.

   A diagonalizable matrix A has a spectral decomposition $A = \sum_{i=1}^{n} \lambda_i |\varphi_i\rangle \langle\varphi_i|$.

Hence, one can define the matrix-to-matrix generalization of f as

$$f_m(A) = \sum_i f(\lambda_i) \left|\varphi_i\right\rangle \left\langle\varphi_i\right| . \tag{24}$$

(Note that this definition of a general matrix function is consistent with the usual exponentiation.)

3. Example: take a scalar $\theta \in \mathbb{C}$. Then,

$$e^{\theta Z} = \begin{bmatrix} e^{\theta} & 0 \\ 0 & e^{-\theta} \end{bmatrix} . \tag{25}$$

This follows from the fact that the spectral decomposition of Z is

$$Z = (+1) \left|\varphi_1\right\rangle \left\langle\varphi_1\right| + (-1) \left|\varphi_2\right\rangle \left\langle\varphi_2\right| , \tag{26}$$

with $\left|\varphi_1\right\rangle$ and $\left|\varphi_2\right\rangle$ being the canonical basis vectors of $\mathbb{C}^2$.

4. Exercise (easy): Find the square root and the logarithm of the matrix

$$A = \begin{bmatrix} 4 & 3 \\ 3 & 4 \end{bmatrix} . \tag{27}$$

5. Exercise (advanced): Let $\vec{v}$ be a real, three-dimensional unit vector, and θ a real number. Prove that

$$\exp\left(i\theta \vec{v} \cdot \vec{\sigma}\right) = \cos(\theta) I + i \sin(\theta) \vec{v} \cdot \vec{\sigma}, \tag{28}$$

where $\vec{v} \cdot \vec{\sigma} = \sum_{i=3}^{3} v_i \sigma_i$.